CAPSTONE PROJECT REPORT

TRENDSTORE CLOUD-NATIVE INFRASTRUCTURE

MARCH 2026

Project Name: TrendStore Cloud-Native Infrastructure

Prepared by: Bibek Kumar Sahu

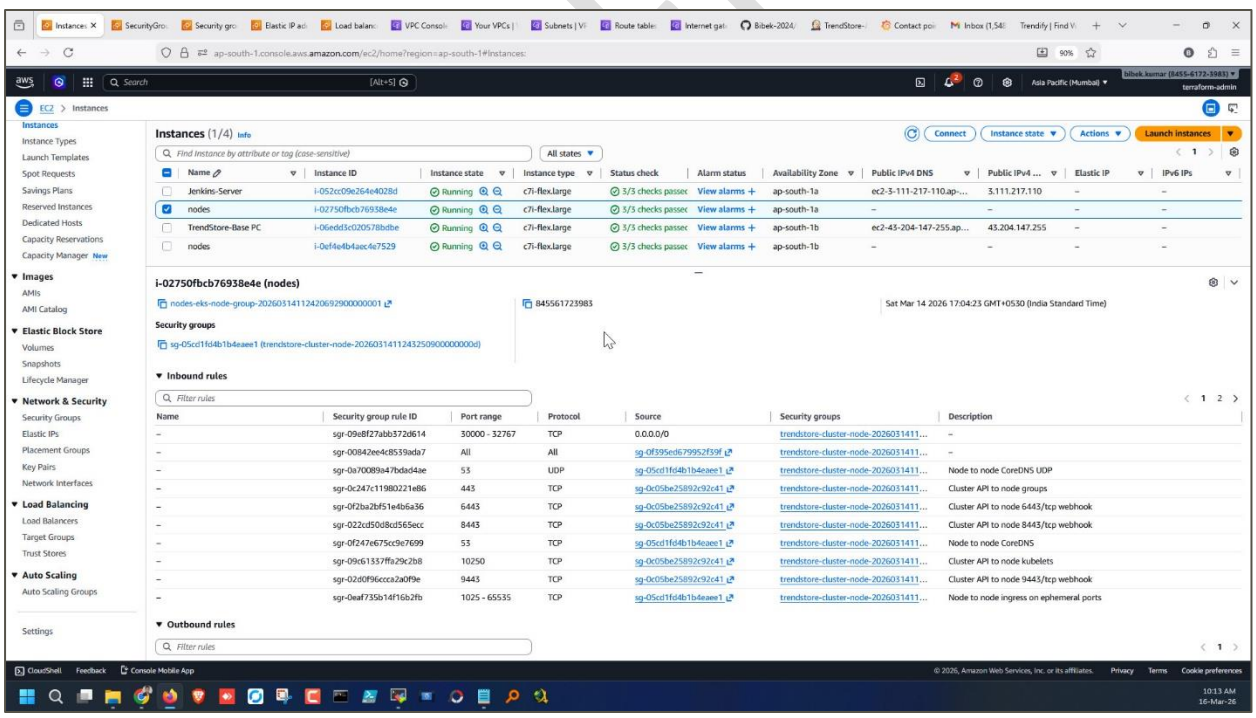
Status: Production-Ready & Monitored

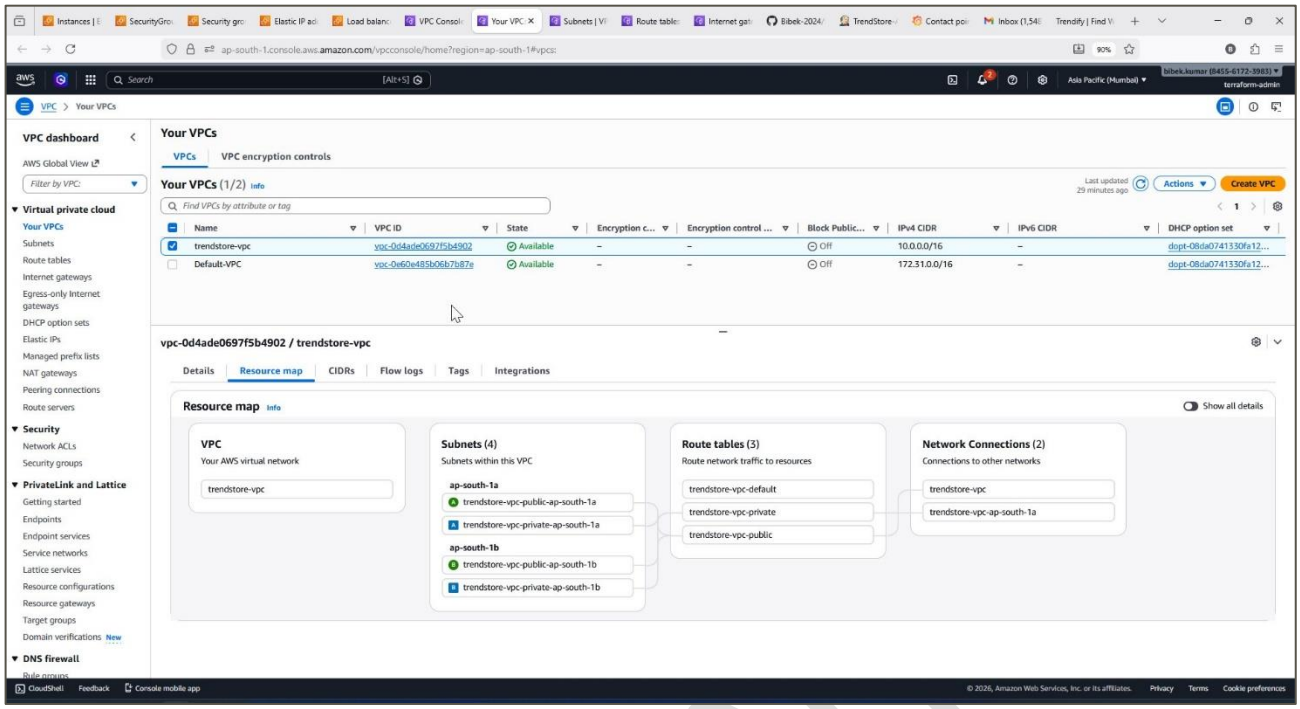
Technology Stack: AWS (EKS, VPC, EC2), Terraform, Jenkins, Docker, Kubernetes, Prometheus, Grafana.

I. Infrastructure-as-Code & Cloud Foundation

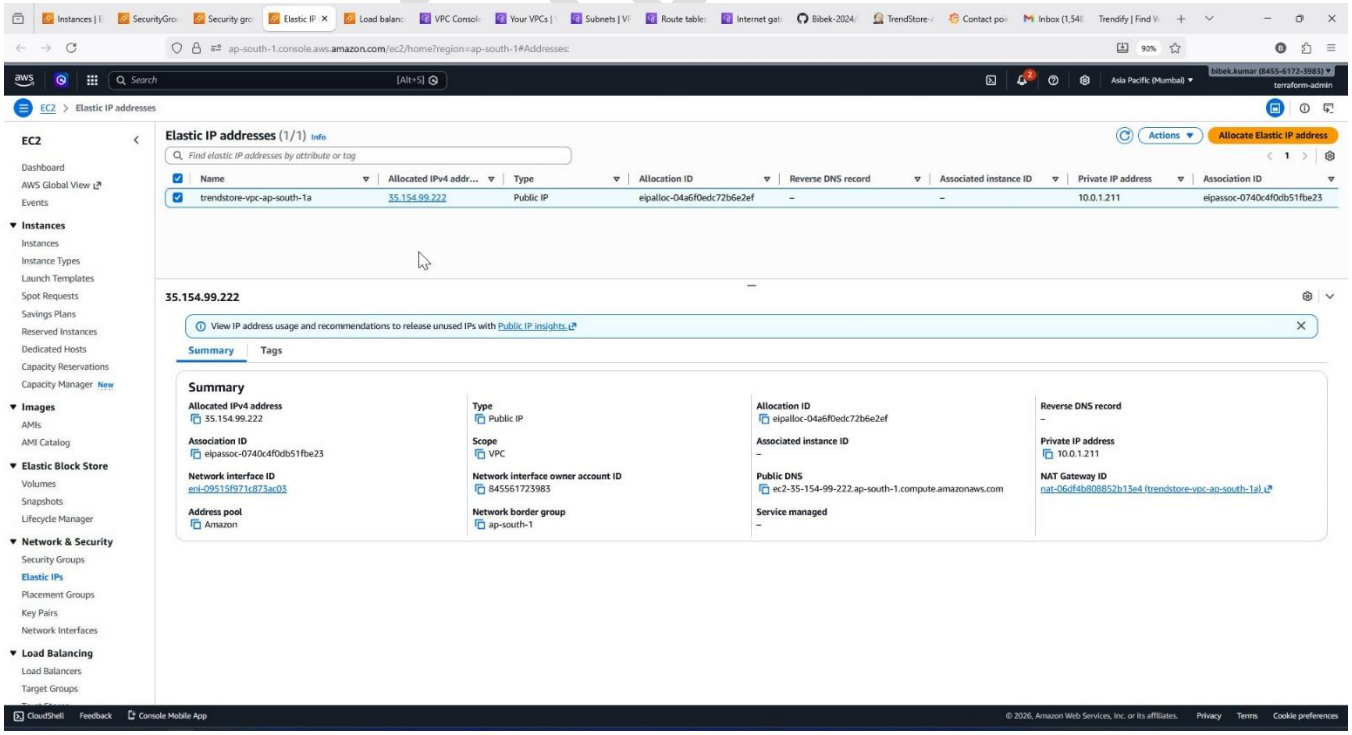
The core networking and compute layers were provisioned using Terraform to ensure a repeatable and secure environment.

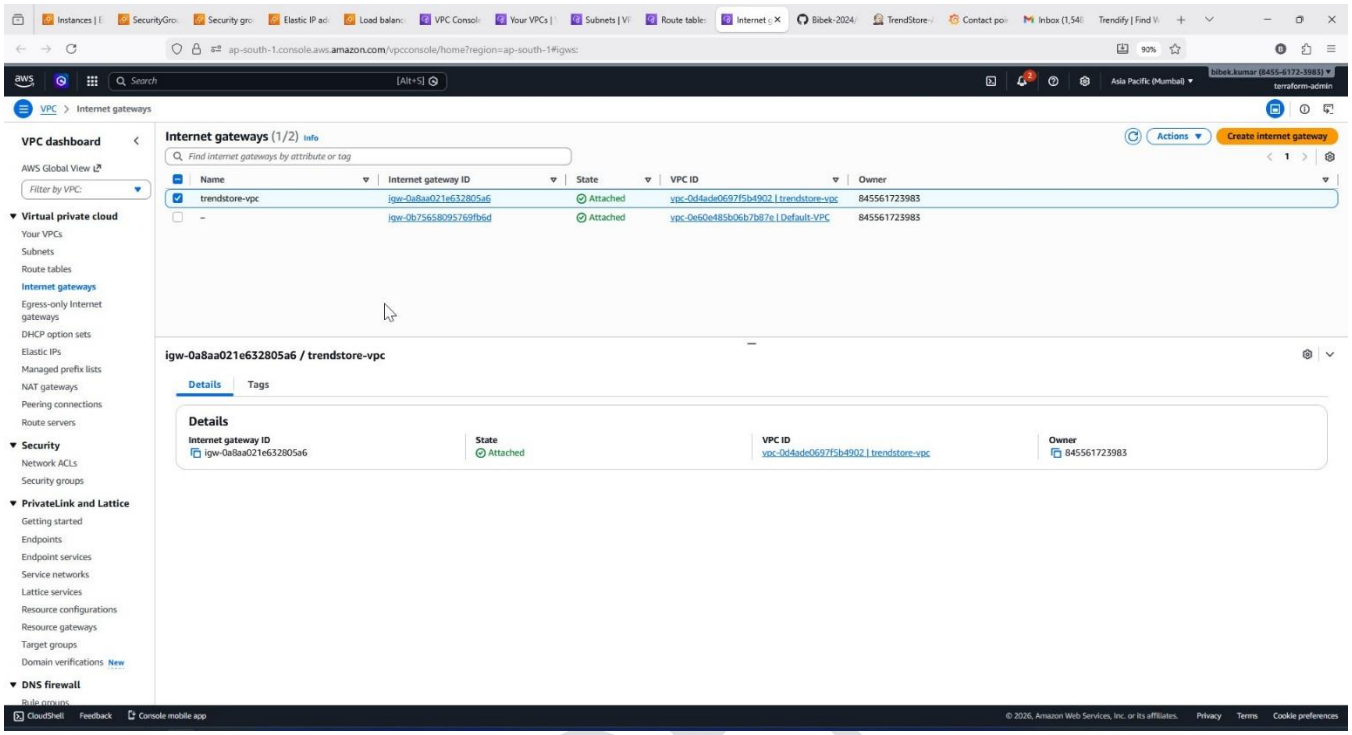
- **[1.1 - 1.3] Compute & Network Architecture:** Deployment of the EKS Cluster within a custom VPC. The resource map confirms high availability across multiple availability zones.

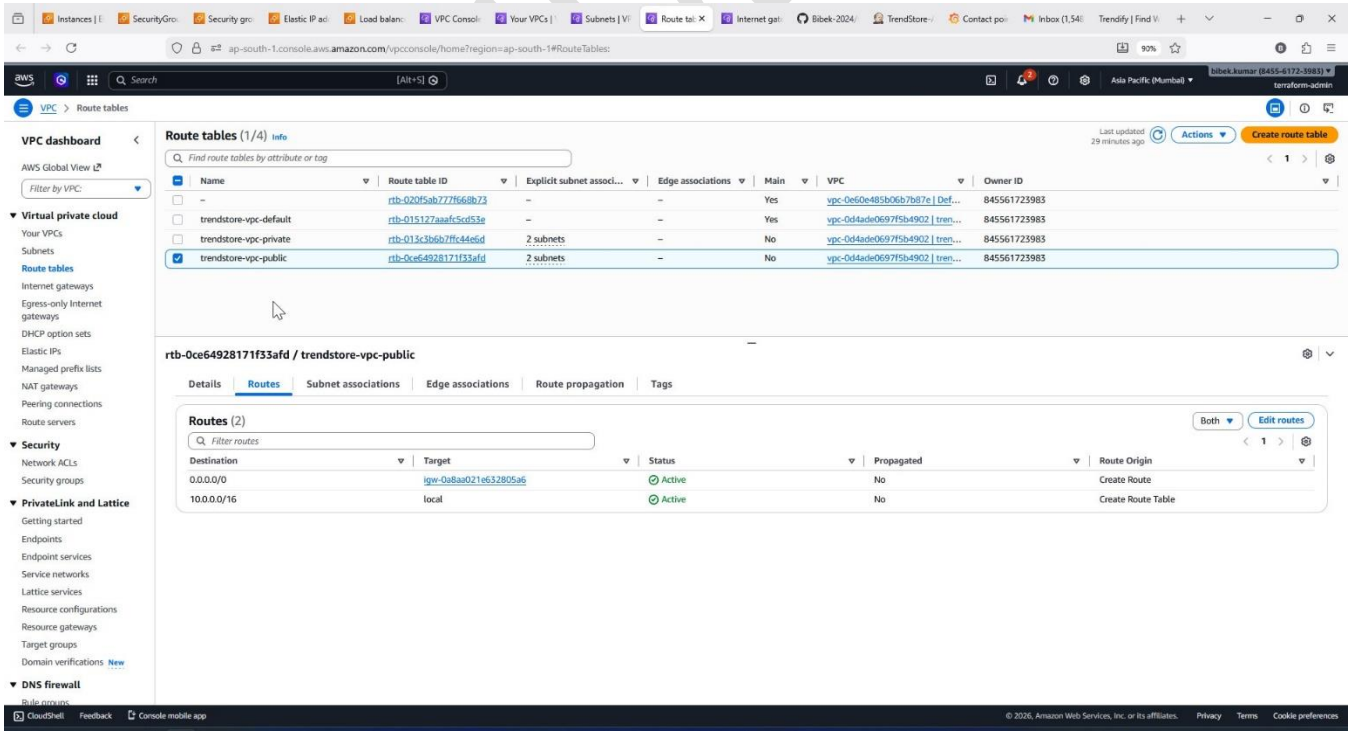


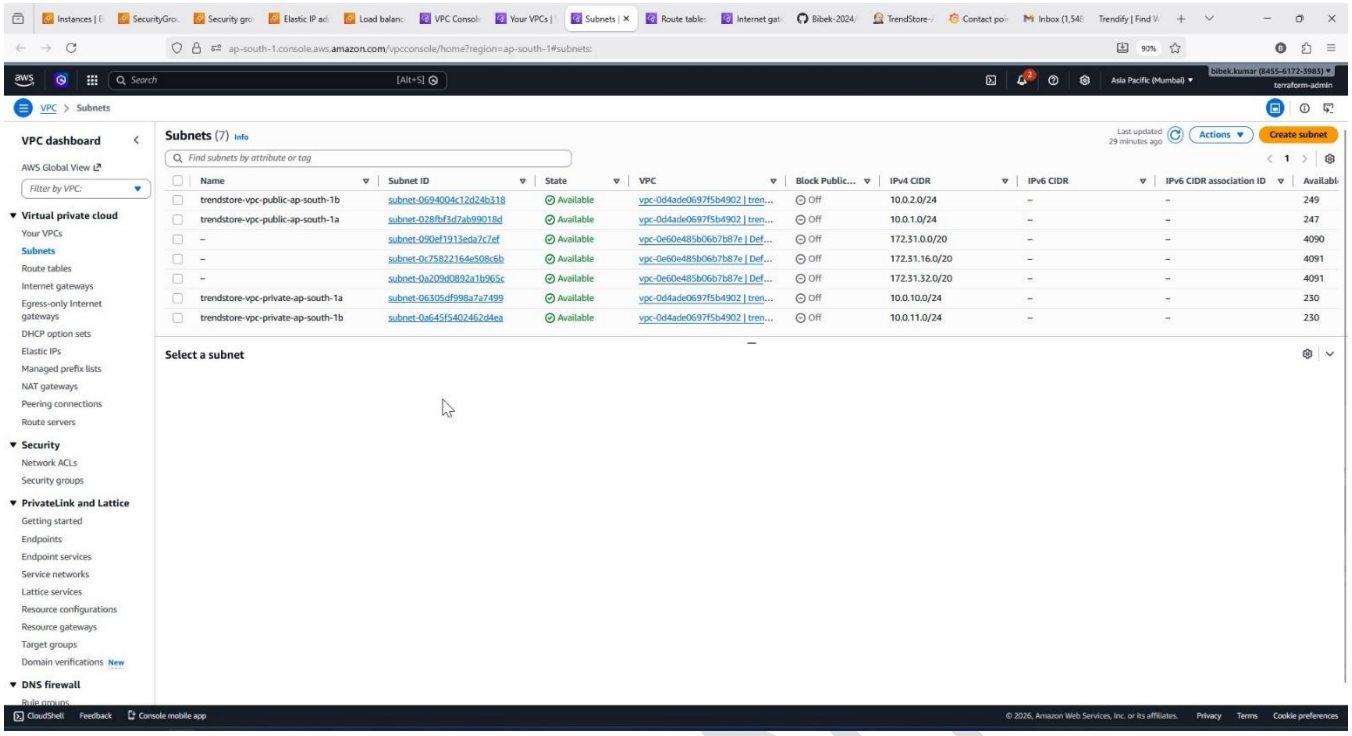


- **[1.4 - 1.7] Traffic Management:** Implementation of NAT Gateways and Elastic IPs to ensure secure outbound traffic, with segmented Public and Private subnets managed by dedicated Route Tables.

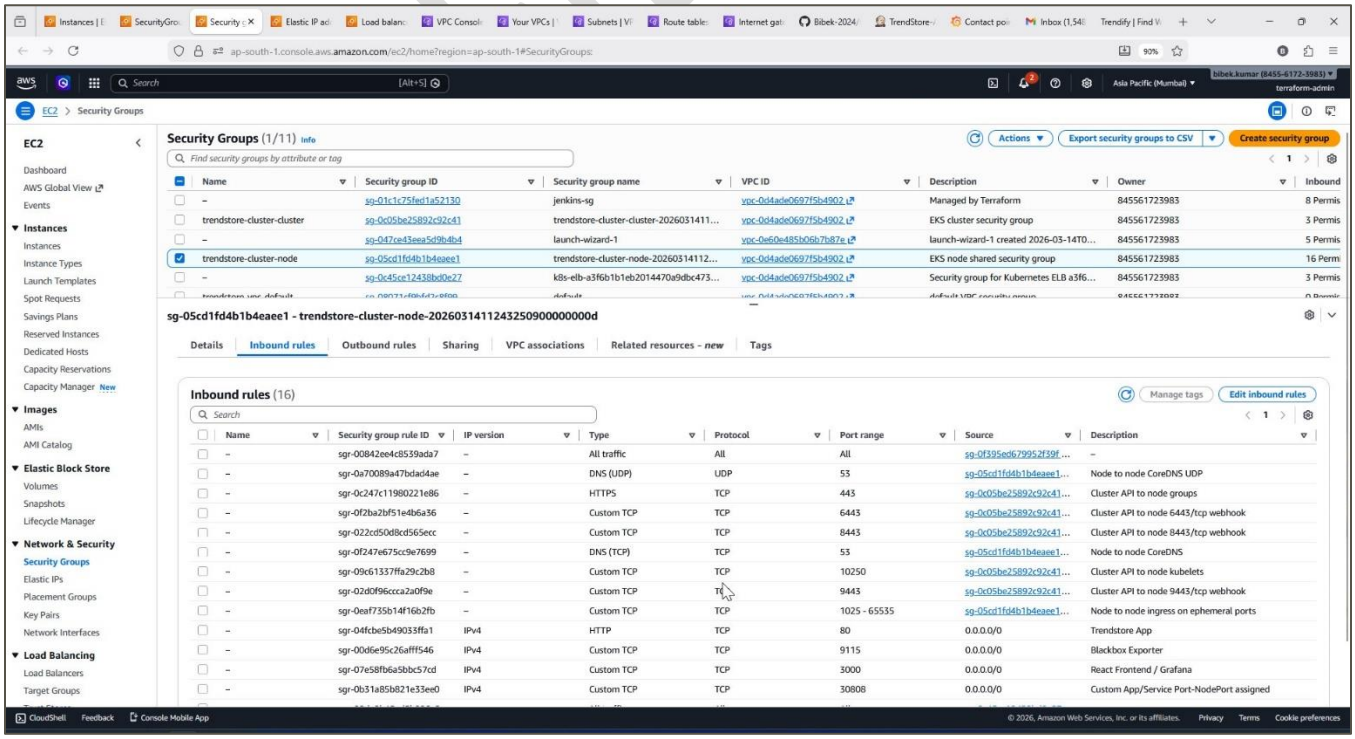



VPC > Internet gateways
VPC dashboard
Internet gateways (1/2) Info
Actions
Create internet gateway
Name
Internet gateway ID
State
VPC ID
Owner
trendstore-vpc
igw-0a8aa021e632805a6
Attached
vpc-0d4ade0697f5b4902 | trendstore-vpc
845561723983
igw-0b75658095769fb6d
Attached
vpc-0e60e485b06b7b87e | Default-VPC
845561723983
igw-0a8aa021e632805a6 / trendstore-vpc
Details
Tags
Internet gateway ID
igw-0a8aa021e632805a6
State
Attached
VPC ID
vpc-0d4ade0697f5b4902 | trendstore-vpc
Owner
845561723983


VPC > Route tables
VPC dashboard
Route tables (1/4) Info
Actions
Create route table
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
Owner ID
rtb-020f5ab777f668b73
Yes
845561723983
trendstore-vpc-default
rtb-015127aaafc5cd53e
Yes
845561723983
trendstore-vpc-private
rtb-013c3b6b7ffc44e6d
2 subnets
No
845561723983
trendstore-vpc-public
rtb-0ce64928171f33afd
2 subnets
No
845561723983
rtb-0ce64928171f33afd / trendstore-vpc-public
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Routes (2)
Both
Edit routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
igw-0a8aa021e632805a6
Active
No
Create Route
10.0.0.0/16
local
Active
No
Create Route Table



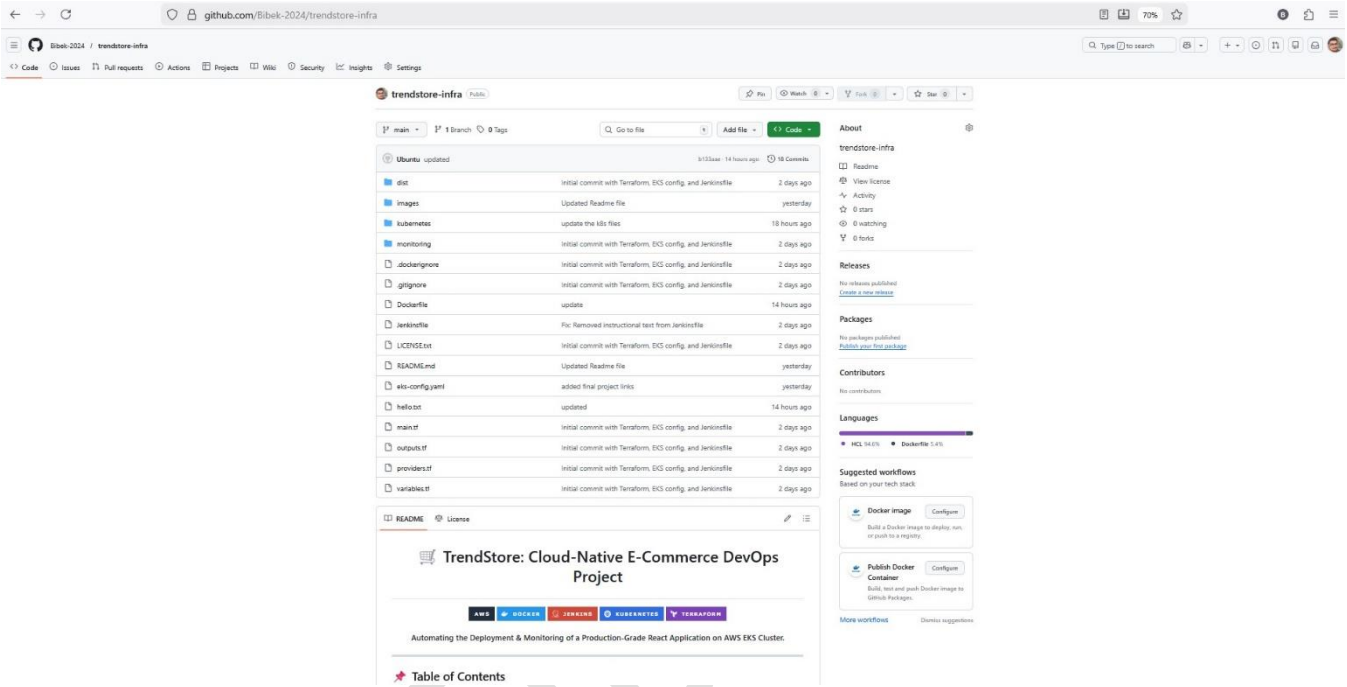
- **[1.2] Security Hardening:** Inbound Security Group rules are strictly defined to allow only necessary traffic (Port 80 for Web, 3000 for Monitoring, and 6443 for K8s API).

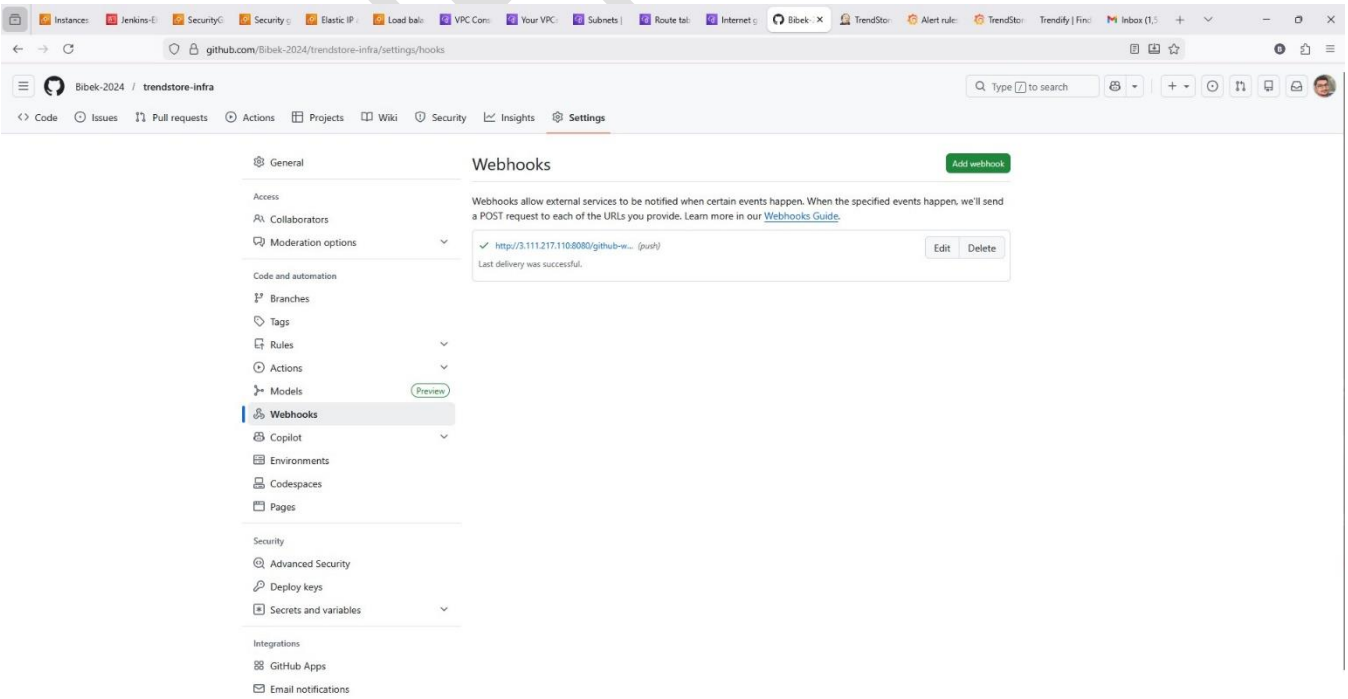


II. Automated CI/CD Lifecycle

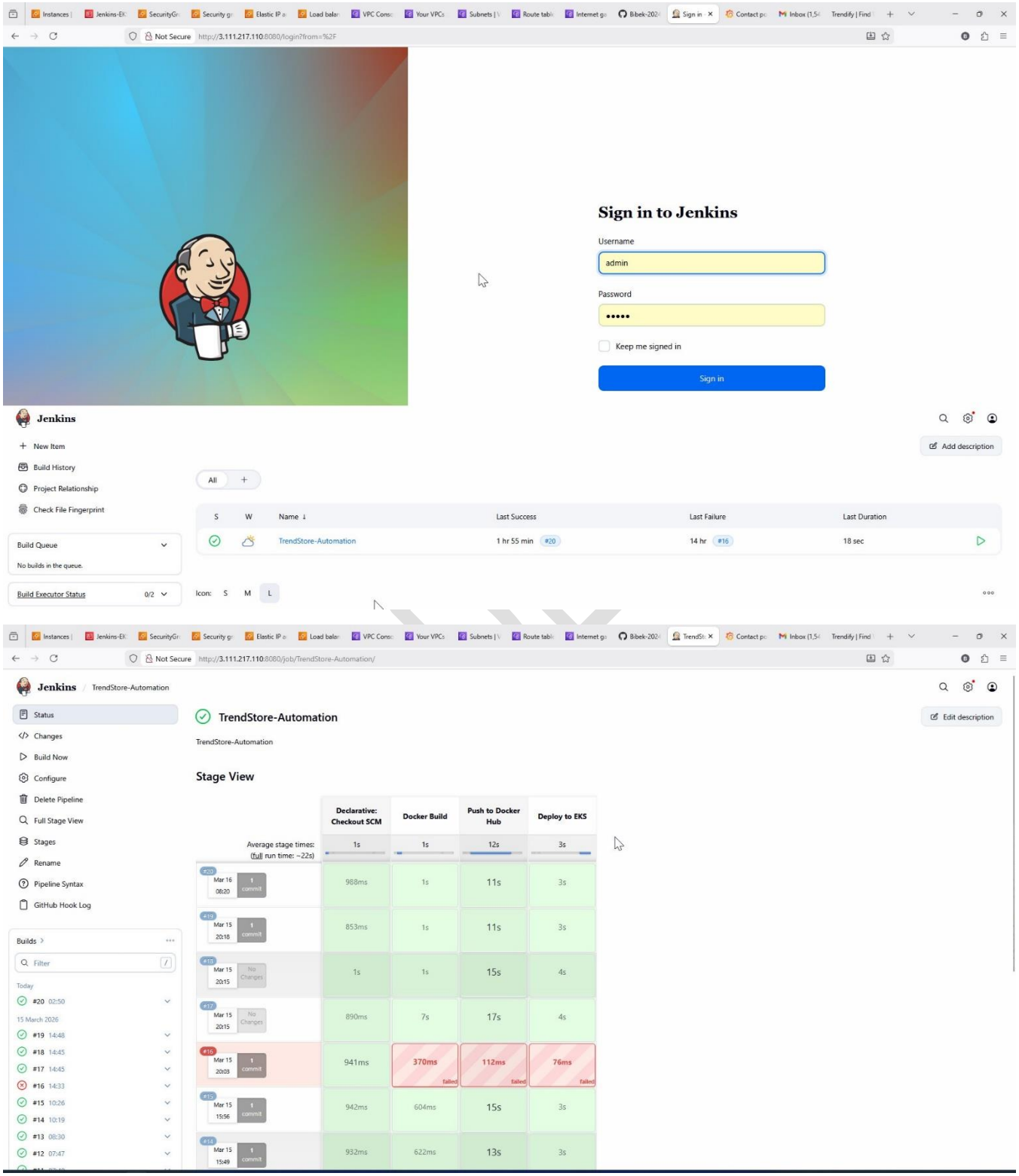
We achieved a "Zero-Touch" deployment strategy where code changes move from GitHub to Production automatically.

- **[2.1 - 2.2] Source Control & Webhooks:** The GitHub repository is integrated with Jenkins via Webhooks, triggering the pipeline on every git push.

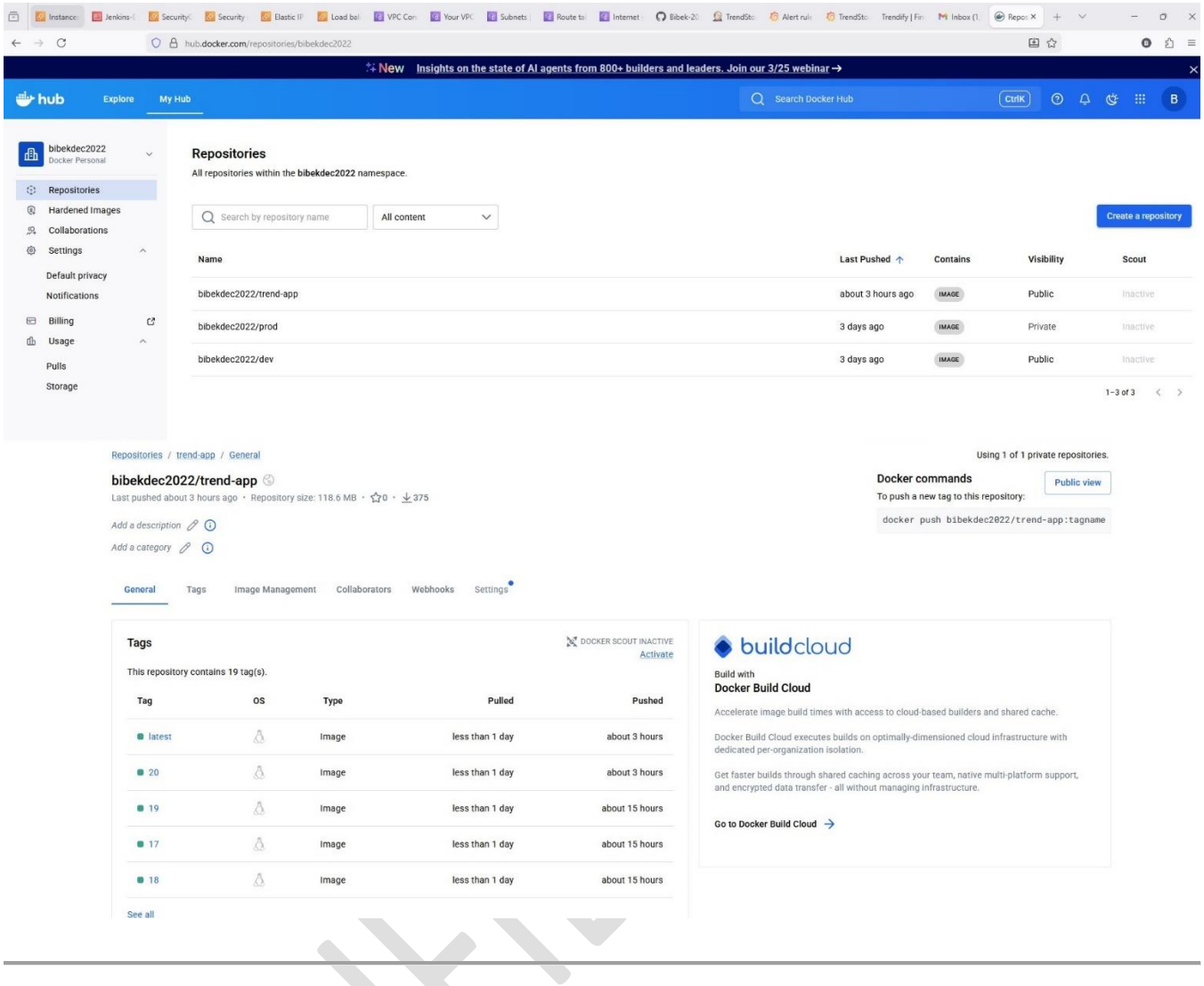


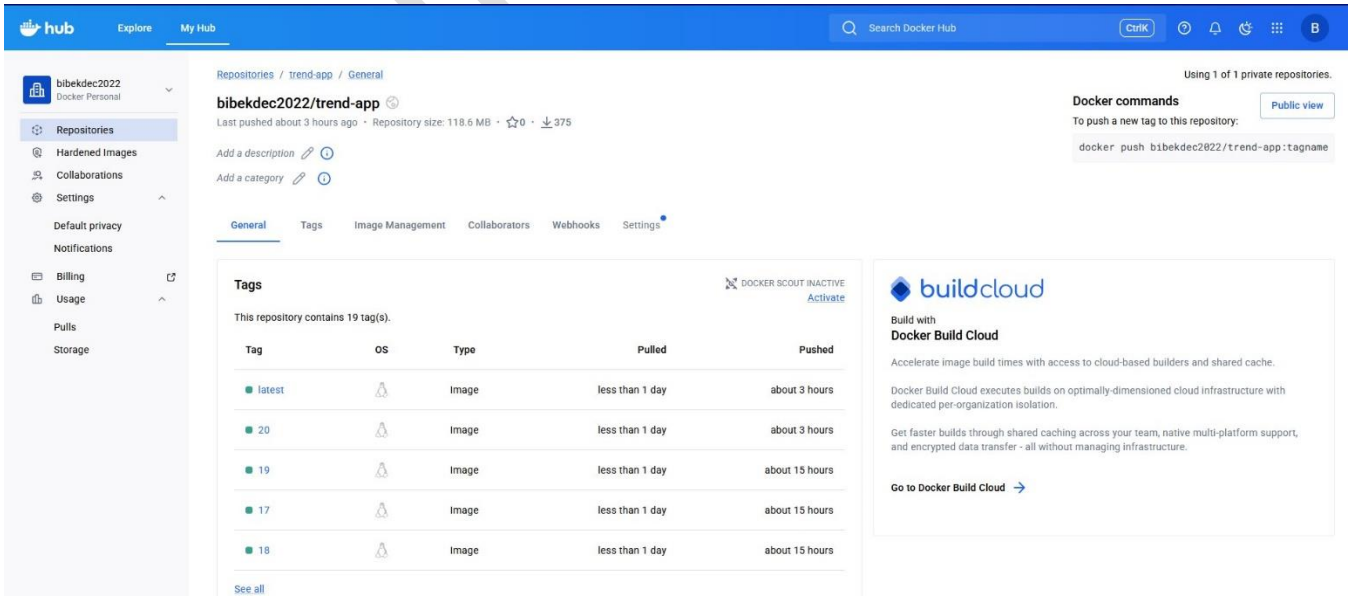


- **[3.1 - 3.2] Jenkins Orchestration:** The pipeline automates code checkout, Docker image creation, and EKS deployment. The stage view provides real-time visibility into build health.



- **[4.0 - 4.2] Artifact Management:** Successful Docker images are versioned and pushed to Docker Hub, serving as a single source of truth for our containers.


hub.docker.com/repositories/bibekdec2022
New Insights on the state of AI agents from 800+ builders and leaders. Join our 3/25 webinar →
hub
Explore
My Hub
Search Docker Hub
bibekdec2022
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Repositories
All repositories within the bibekdec2022 namespace.
Search by repository name
All content
Create a repository
Name
Last Pushed
Contains
Visibility
Scout
bibekdec2022/trend-app
about 3 hours ago
IMAGE
Public
Inactive
bibekdec2022/prod
3 days ago
IMAGE
Private
Inactive
bibekdec2022/dev
3 days ago
IMAGE
Public
Inactive
1–3 of 3
Repositories / trend-app / General
Using 1 of 1 private repositories.
bibekdec2022/trend-app
Last pushed about 3 hours ago • Repository size: 118.6 MB • 0 • 375
Add a description
Add a category
Docker commands
Public view
To push a new tag to this repository:
docker push bibekdec2022/trend-app:tagname
General
Tags
Image Management
Collaborators
Webhooks
Settings
Tags
DOCKER SCOUT INACTIVE
Activate
This repository contains 19 tag(s).
Tag
OS
Type
Pulled
Pushed
latest
Image
less than 1 day
about 3 hours
20
Image
less than 1 day
about 3 hours
19
Image
less than 1 day
about 15 hours
17
Image
less than 1 day
about 15 hours
18
Image
less than 1 day
about 15 hours
See all
buildcloud
Build with
Docker Build Cloud
Accelerate image build times with access to cloud-based builders and shared cache.
Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.
Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.
Go to Docker Build Cloud


hub
Explore
My Hub
Search Docker Hub
bibekdec2022
Docker Personal
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Repositories / trend-app / General
Using 1 of 1 private repositories.
bibekdec2022/trend-app
Last pushed about 3 hours ago • Repository size: 118.6 MB • 0 • 375
Add a description
Add a category
Docker commands
Public view
To push a new tag to this repository:
docker push bibekdec2022/trend-app:tagname
General
Tags
Image Management
Collaborators
Webhooks
Settings
Tags
DOCKER SCOUT INACTIVE
Activate
This repository contains 19 tag(s).
Tag
OS
Type
Pulled
Pushed
latest
Image
less than 1 day
about 3 hours
20
Image
less than 1 day
about 3 hours
19
Image
less than 1 day
about 15 hours
17
Image
less than 1 day
about 15 hours
18
Image
less than 1 day
about 15 hours
See all
buildcloud
Build with
Docker Build Cloud
Accelerate image build times with access to cloud-based builders and shared cache.
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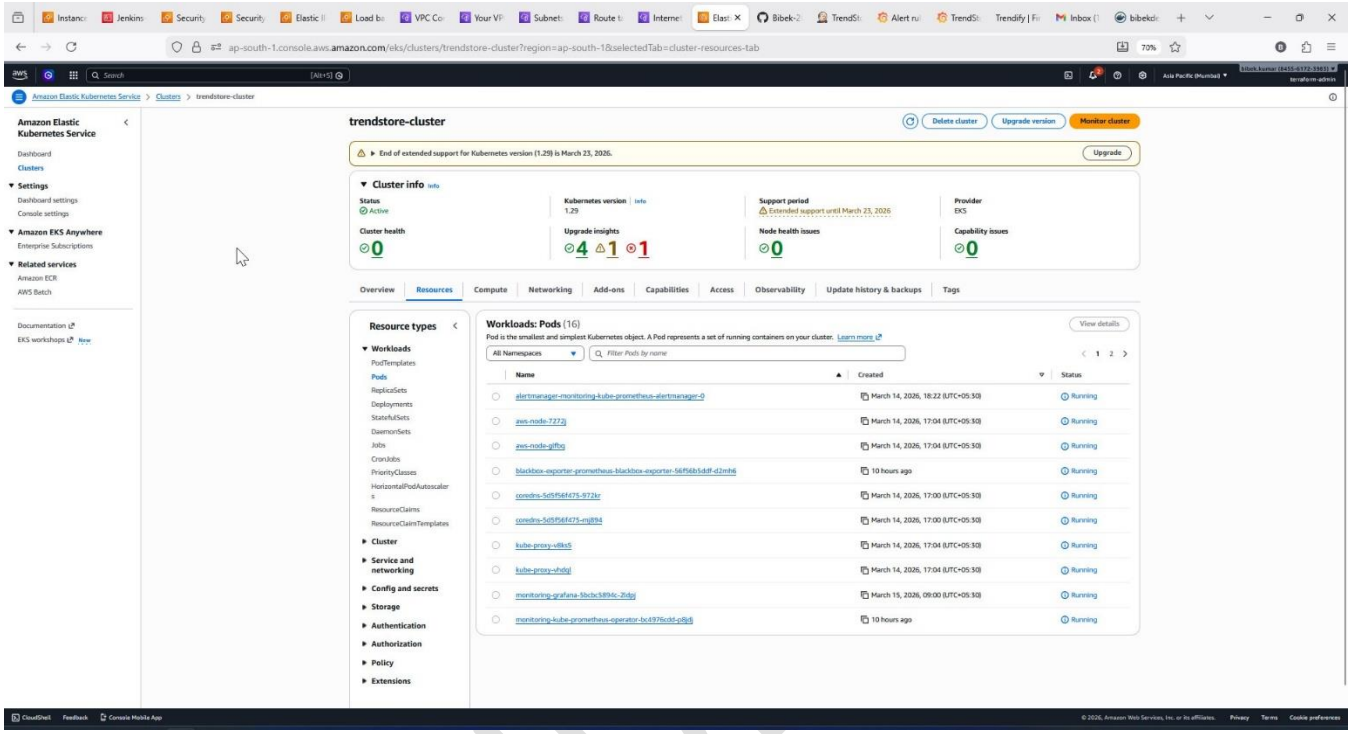
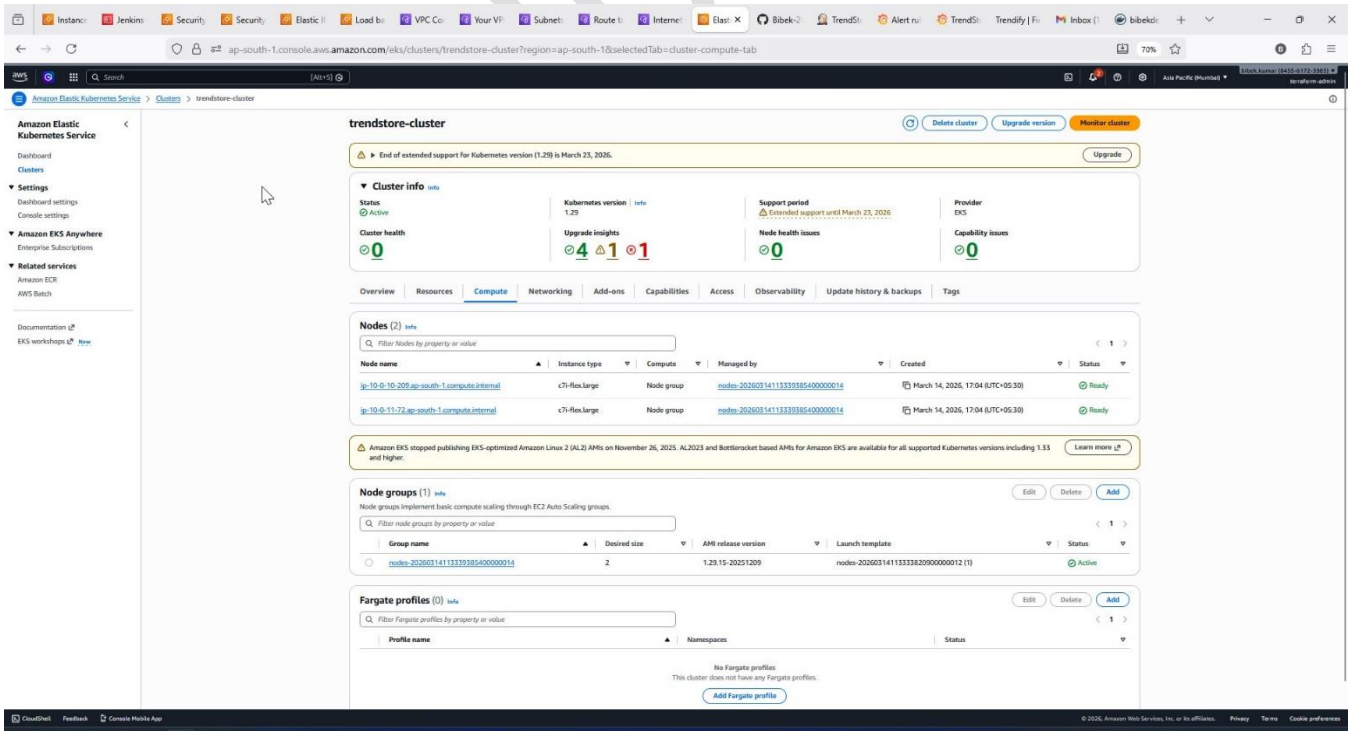
Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.
Go to Docker Build Cloud

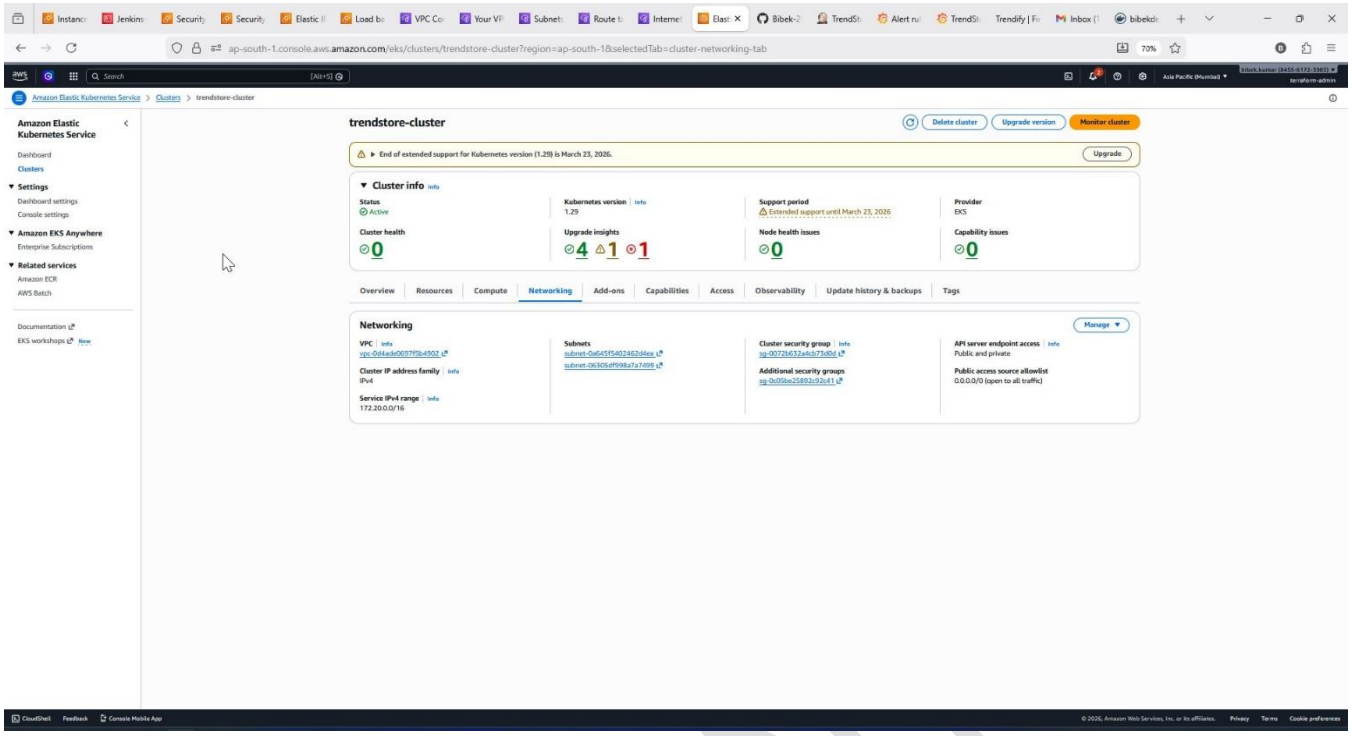
III. Kubernetes Orchestration & Delivery

The application is hosted on AWS EKS for maximum scalability and resilience.

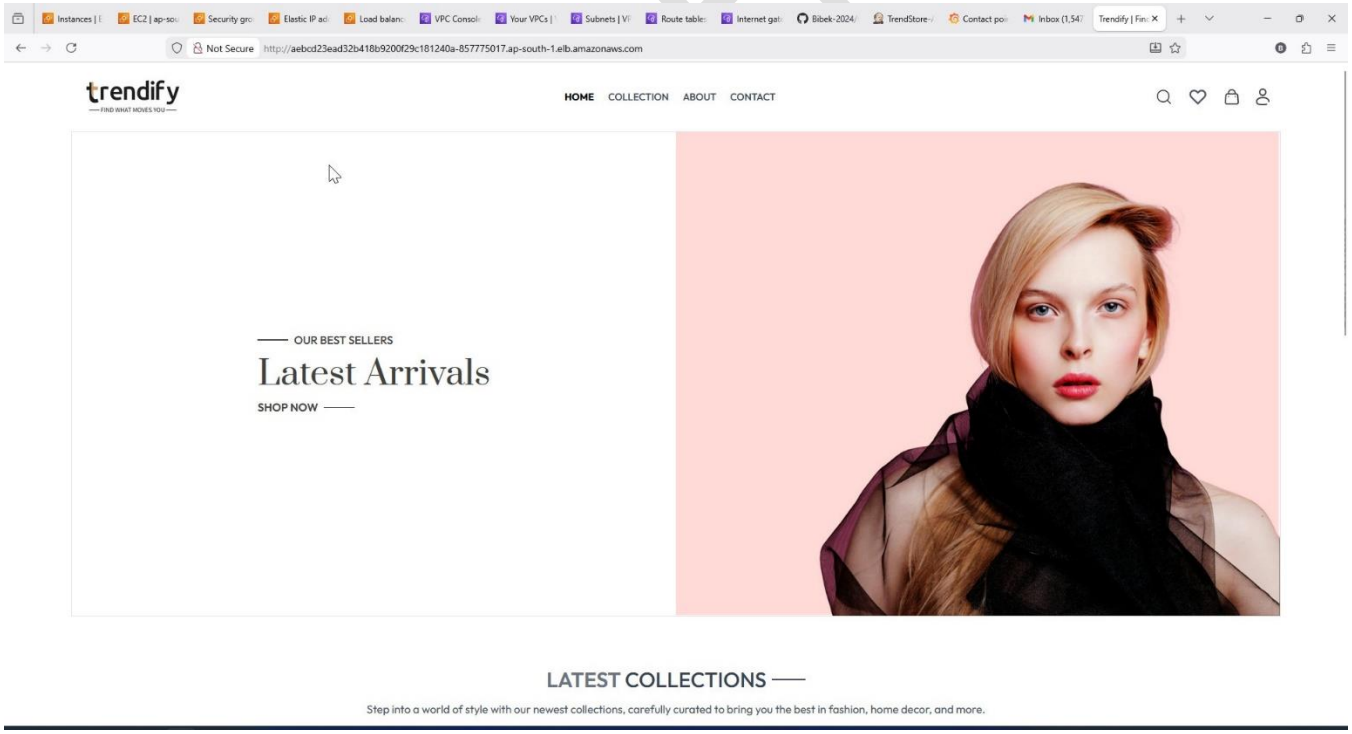
- **[5.1 - 5.3] EKS Cluster Management:** Live verification of cluster workloads, including Node Groups and LoadBalancer services that expose the application to the public internet.

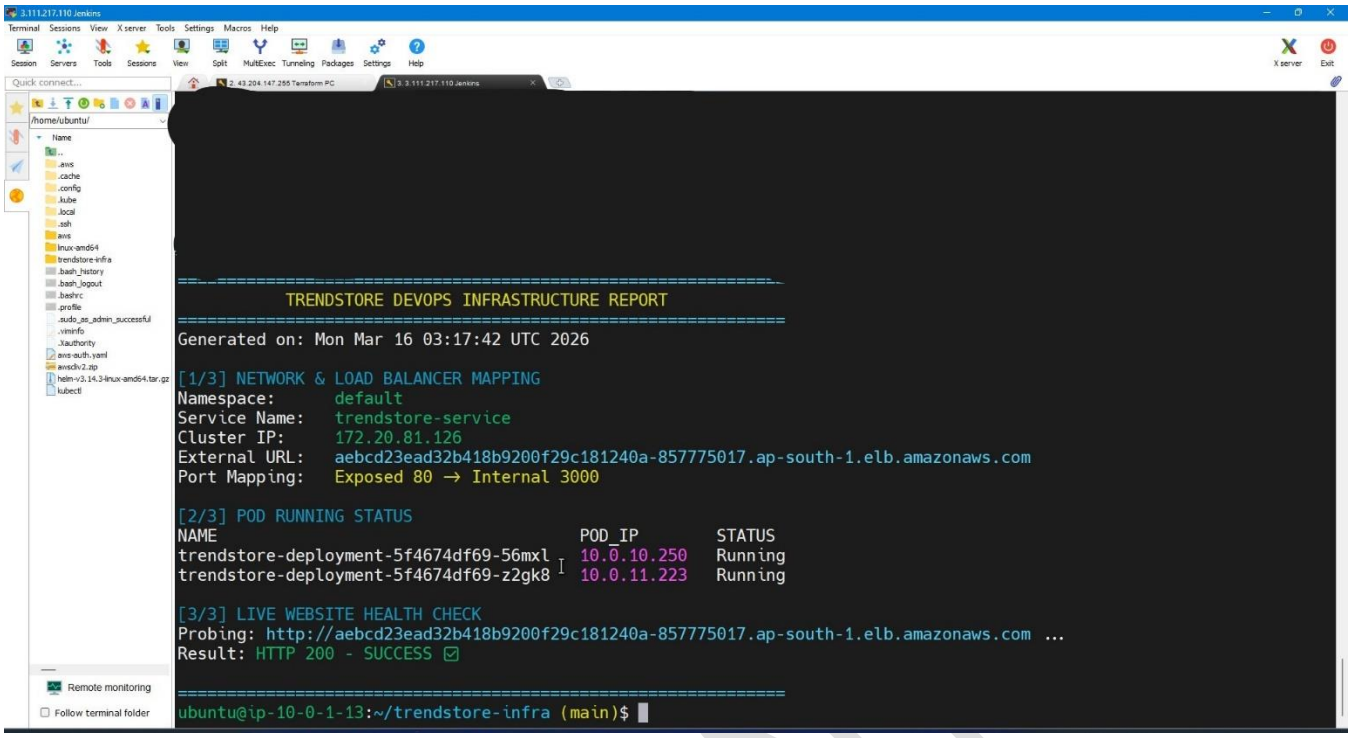






- **[6.1 - 6.2] End-User Verification:** The TrendStore live website is accessible and functional, confirming that the entire infrastructure and application stack are properly aligned.

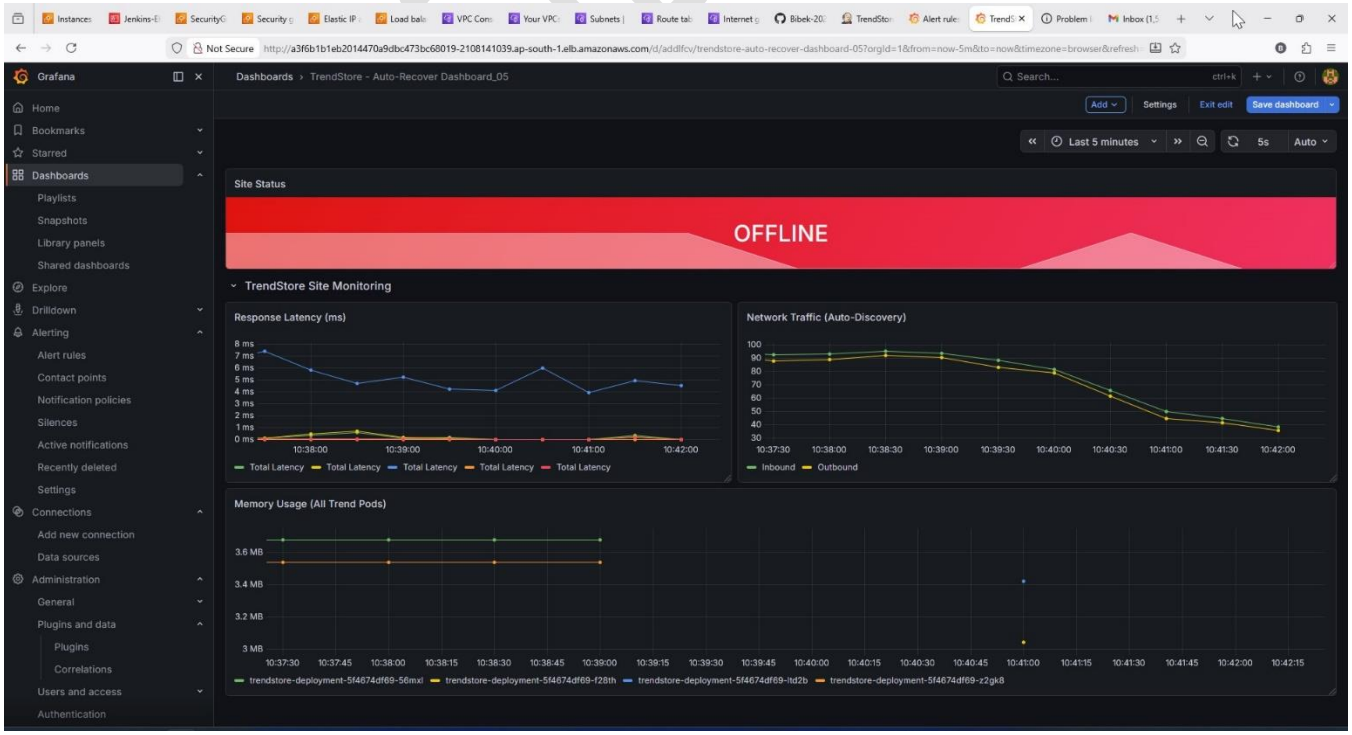


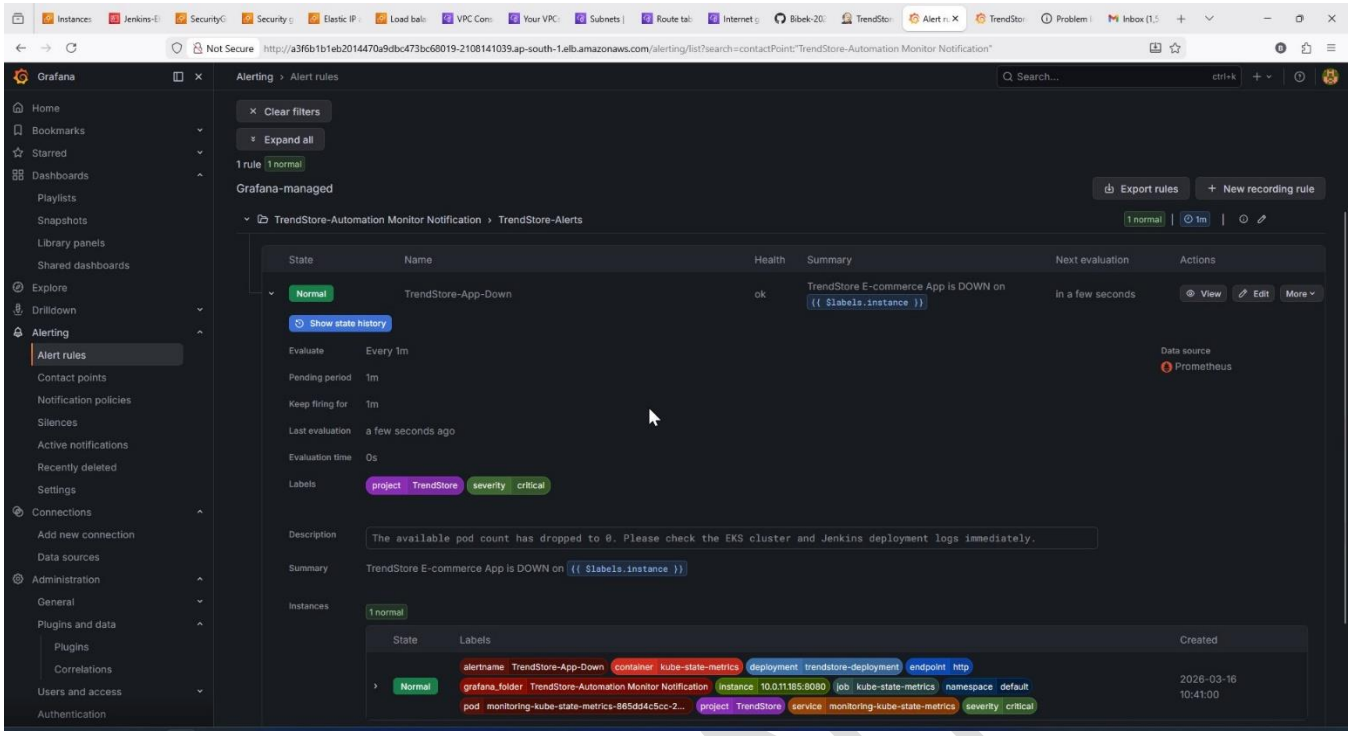


IV. Observability & Automated Incident Response

A robust monitoring stack was implemented to detect, notify, and track system health automatically.

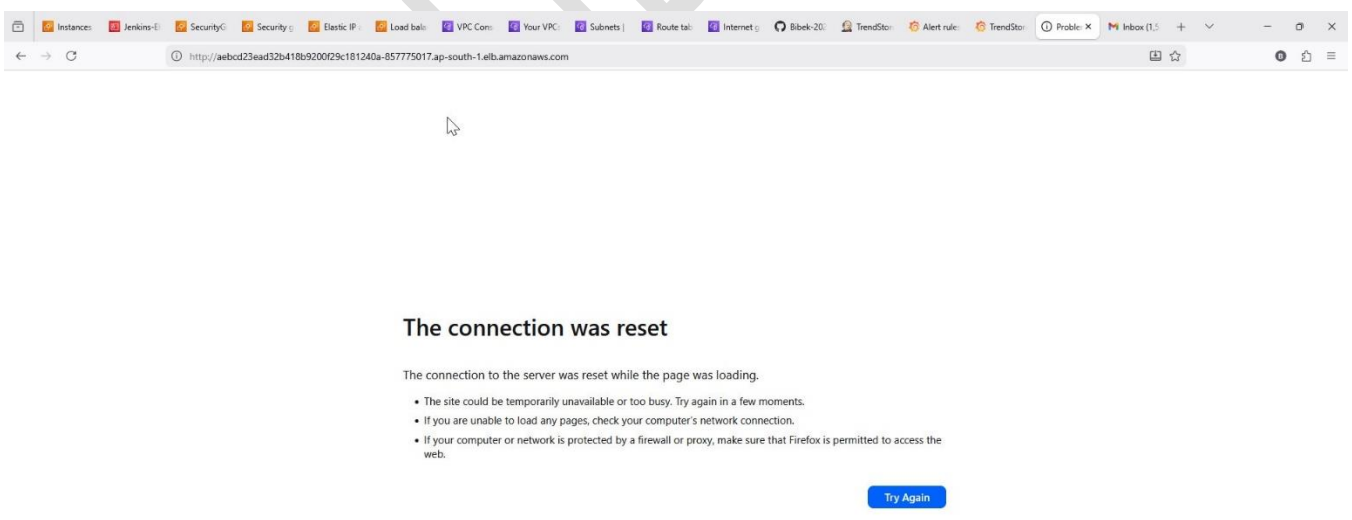
- **[7.2 - 7.3] Full-Stack Monitoring:** Grafana dashboards provide real-time metrics for Memory, Network Traffic, Response Latency and Application Uptime.



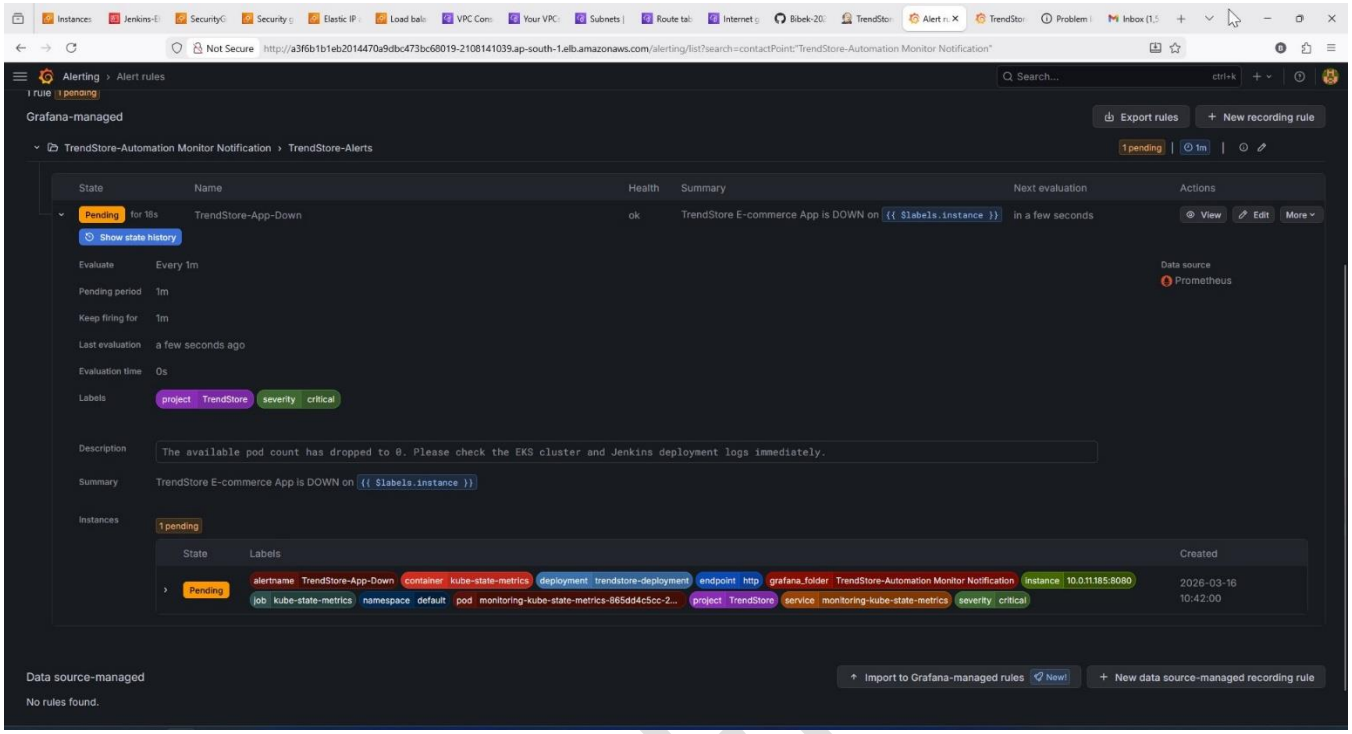


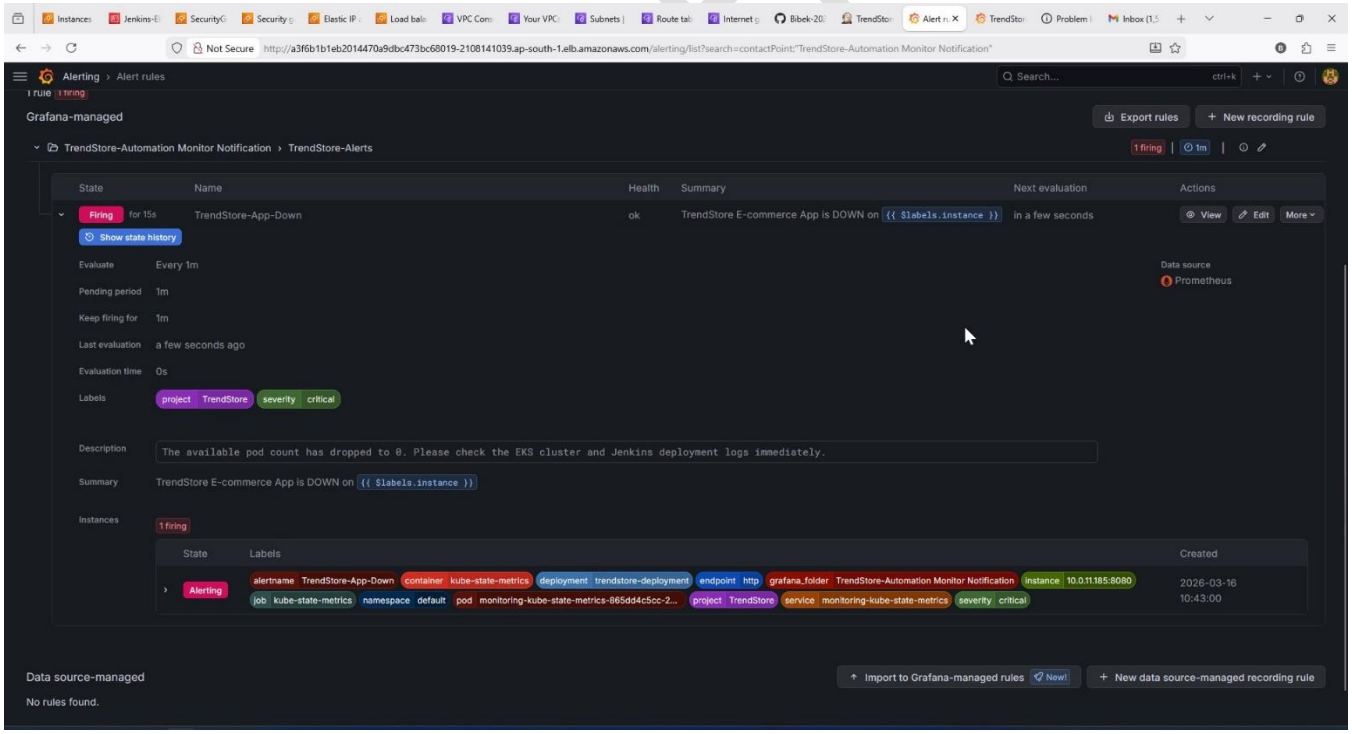
- **[7.0 - 7.1] Disaster Simulation:** Manual scaling of pods to zero was used to verify the sensitivity of our monitoring probes.

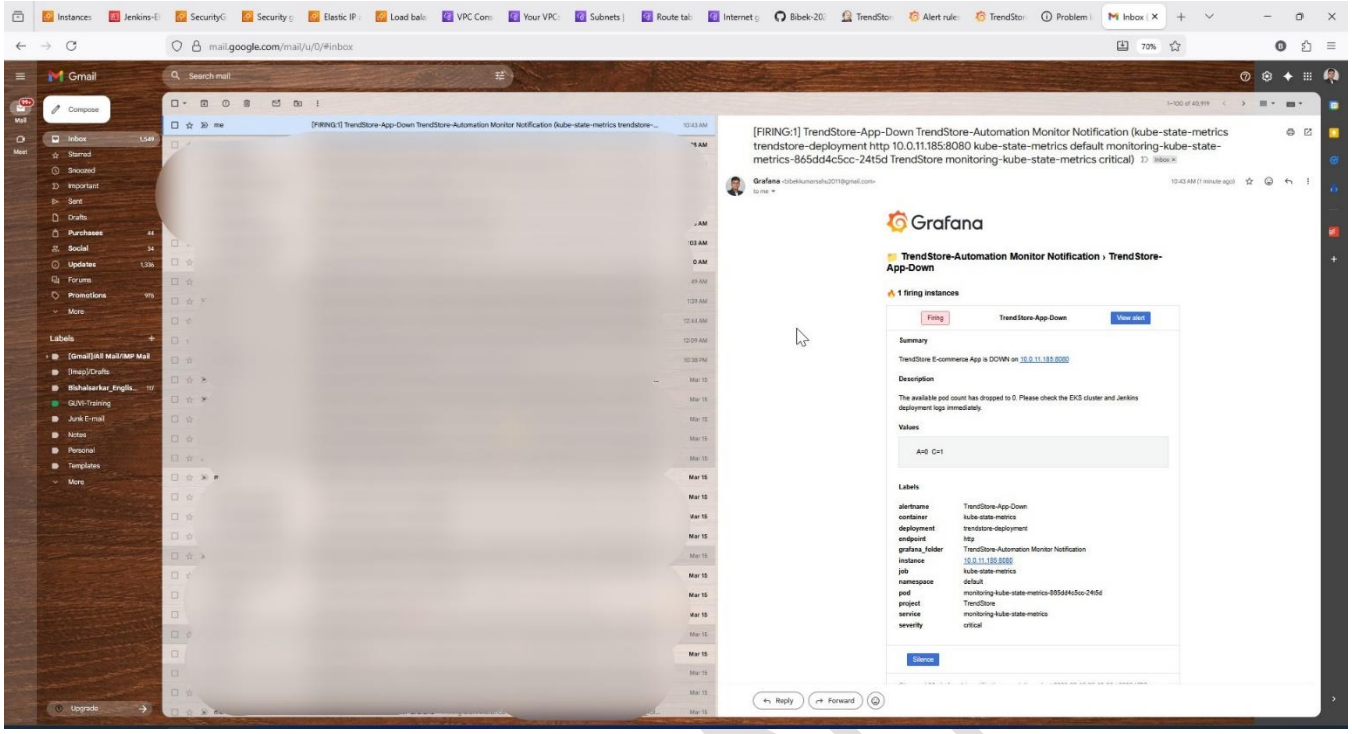
```
ubuntu@ip-10-0-1-13:~/trendstore-infra (main)$ kubectl scale deployment trendstore-deployment --replicas=0 -n default
deployment.apps/trendstore-deployment scaled
ubuntu@ip-10-0-1-13:~/trendstore-infra (main)$ kubectl scale deployment trendstore-deployment --replicas=2 -n default
deployment.apps/trendstore-deployment scaled
ubuntu@ip-10-0-1-13:~/trendstore-infra (main)$
```



- **[7.4 - 7.6] Alerting Engine:** When downtime is detected, the alert state transitions from **Pending** to **Firing**, automatically triggering an SMTP email notification to the DevOps team.

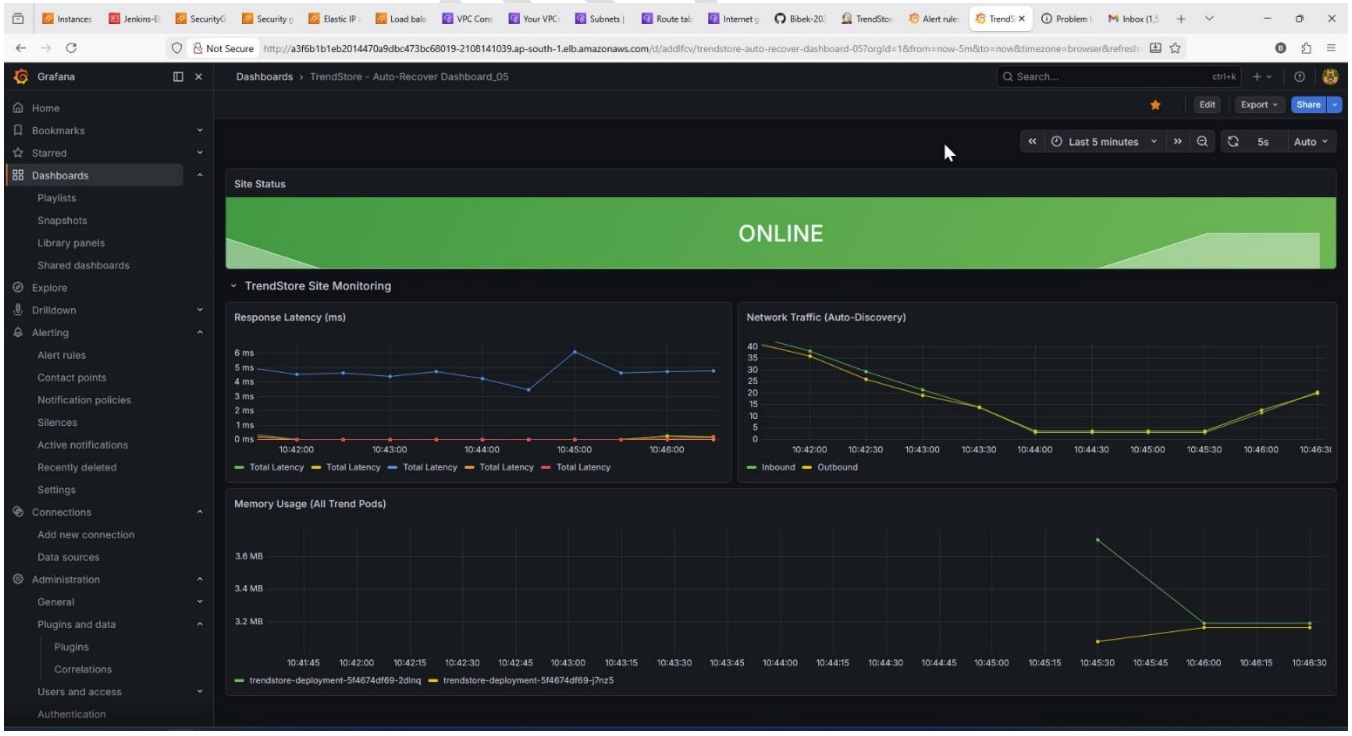


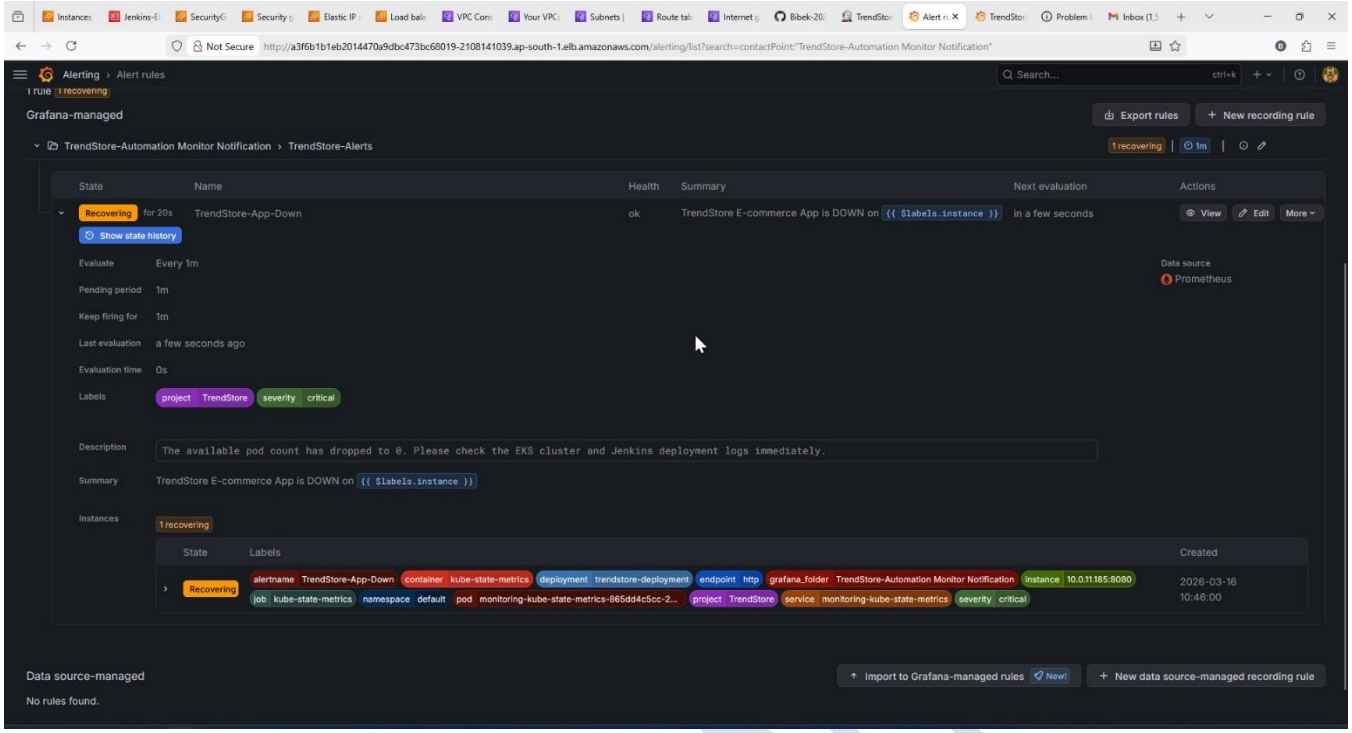


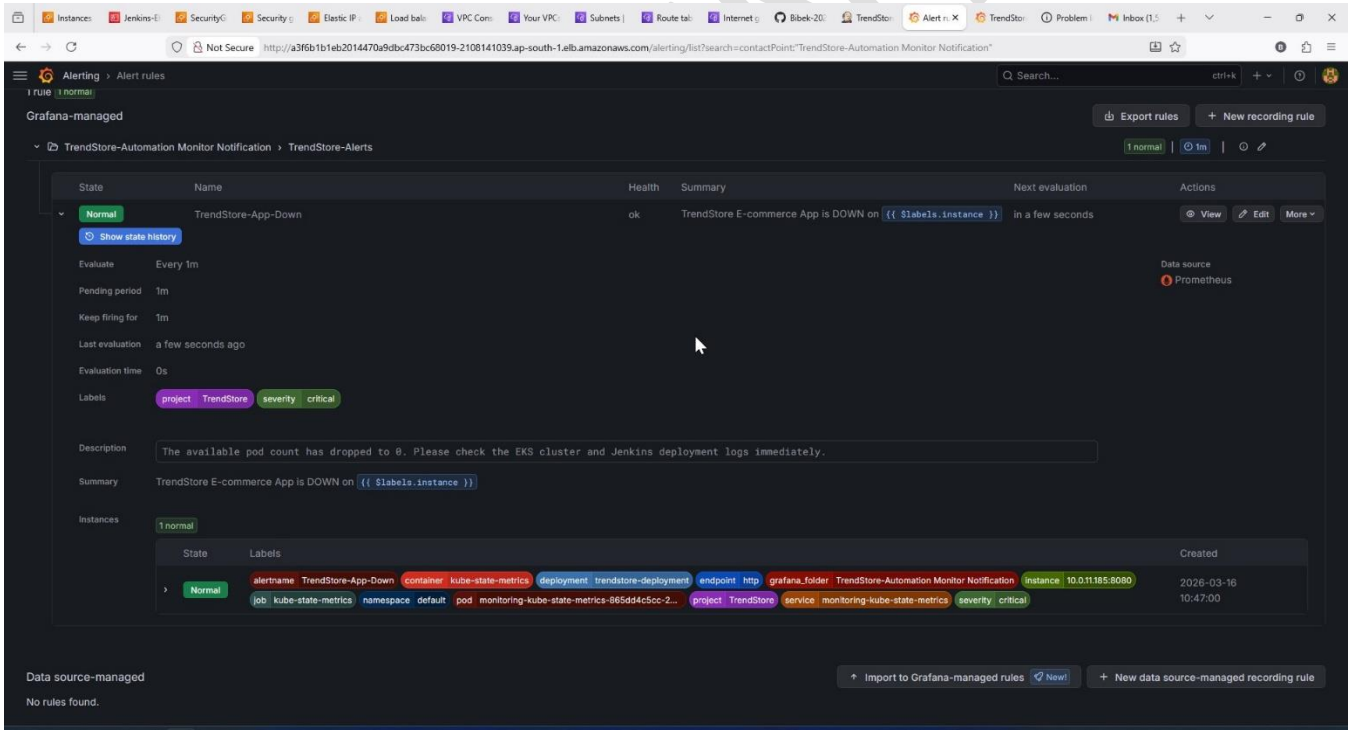


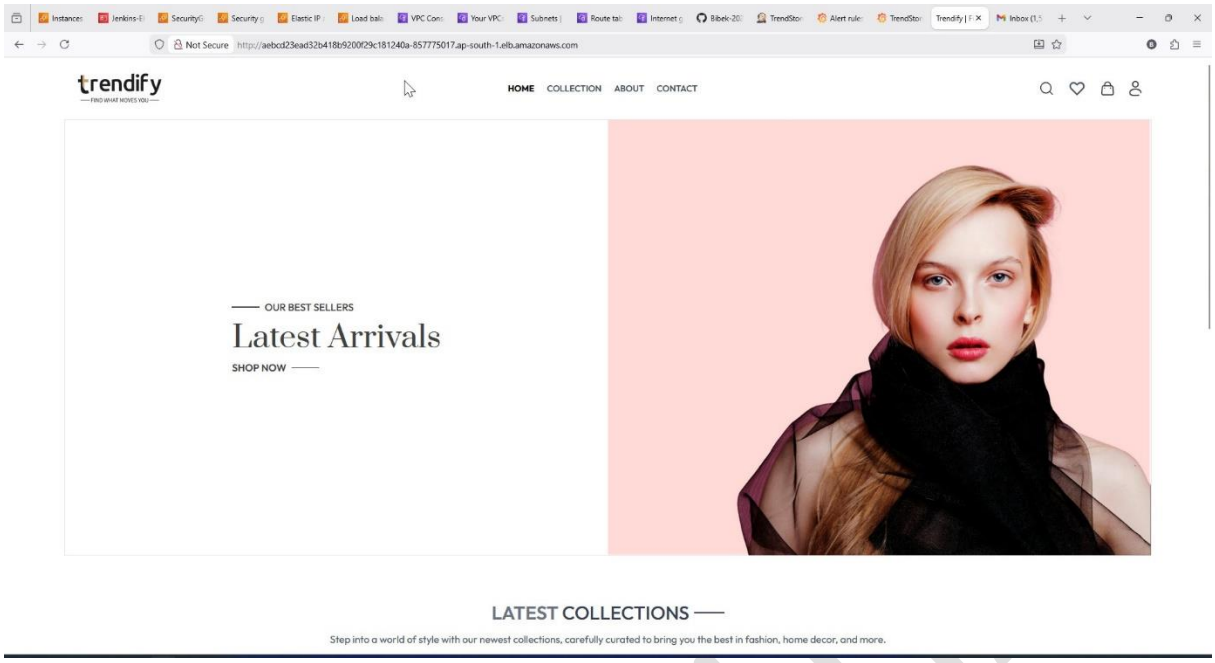
- **[7.7 - 7.11] Service Recovery:** After scaling pods back to the desired state, Grafana reflects a return to **"Online"** status.

```
ubuntu@ip-10-0-1-13:~/trendstore-infra (main)$ kubectl scale deployment trendstore-deployment --replicas=0 -n default
deployment.apps/trendstore-deployment scaled
ubuntu@ip-10-0-1-13:~/trendstore-infra (main)$ kubectl scale deployment trendstore-deployment --replicas=2 -n default
deployment.apps/trendstore-deployment scaled
ubuntu@ip-10-0-1-13:~/trendstore-infra (main)$
```

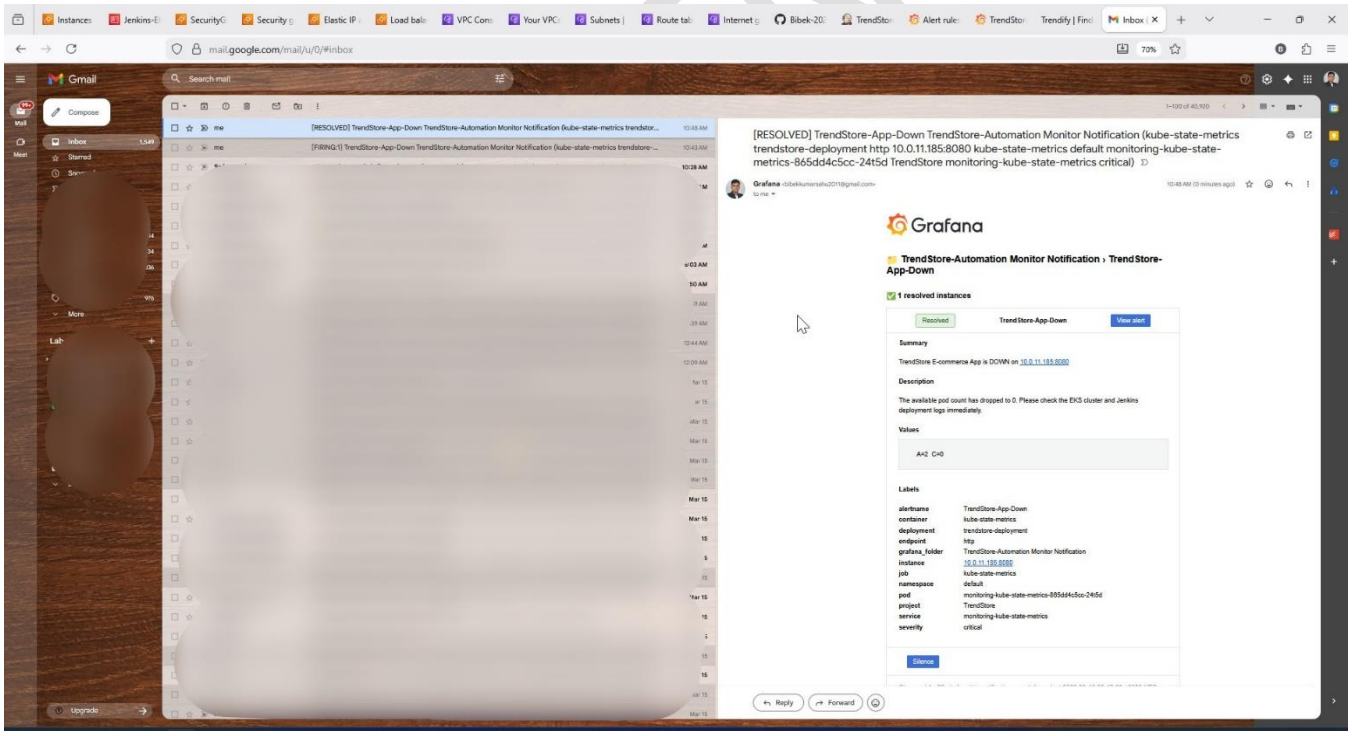



Alerting › Alert rules
1 rule 1 recovering
Grafana-managed
Export rules
+ New recording rule
TrendStore-Automation Monitor Notification › TrendStore-Alerts
1 recovering | 1m
State
Name
Health
Summary
Next evaluation
Actions
Recovering for 20s
TrendStore-App-Down
ok
TrendStore E-commerce App is DOWN on {{ $labels.instance }}
in a few seconds
View
Edit
More
Show state history
Evaluate
Every 1m
Pending period
1m
Keep firing for
1m
Last evaluation
a few seconds ago
Evaluation time
0s
Labels
project TrendStore
severity critical
Data source
Prometheus
Description
The available pod count has dropped to 0. Please check the EKS cluster and Jenkins deployment logs immediately.
Summary
TrendStore E-commerce App is DOWN on {{ $labels.instance }}
Instances
1 recovering
State
Labels
Created
Recovering
alertname TrendStore-App-Down
container kube-state-metrics
deployment trendstore-deployment
endpoint http
grafana_folder TrendStore-Automation Monitor Notification
instance 10.0.11.185:8080
job kube-state-metrics
namespace default
pod monitoring-kube-state-metrics-865dd4c5cc-2...
project TrendStore
service monitoring-kube-state-metrics
severity critical
2026-03-16 10:46:00
Data source-managed
Import to Grafana-managed rules
New!
+ New data source-managed recording rule
No rules found.


Alerting › Alert rules
1 rule 1 normal
Grafana-managed
Export rules
+ New recording rule
TrendStore-Automation Monitor Notification › TrendStore-Alerts
1 normal | 1m
State
Name
Health
Summary
Next evaluation
Actions
Normal
TrendStore-App-Down
ok
TrendStore E-commerce App is DOWN on {{ $labels.instance }}
in a few seconds
View
Edit
More
Show state history
Evaluate
Every 1m
Pending period
1m
Keep firing for
1m
Last evaluation
a few seconds ago
Evaluation time
0s
Labels
project TrendStore
severity critical
Data source
Prometheus
Description
The available pod count has dropped to 0. Please check the EKS cluster and Jenkins deployment logs immediately.
Summary
TrendStore E-commerce App is DOWN on {{ $labels.instance }}
Instances
1 normal
State
Labels
Created
Normal
alertname TrendStore-App-Down
container kube-state-metrics
deployment trendstore-deployment
endpoint http
grafana_folder TrendStore-Automation Monitor Notification
instance 10.0.11.185:8080
job kube-state-metrics
namespace default
pod monitoring-kube-state-metrics-865dd4c5cc-2...
project TrendStore
service monitoring-kube-state-metrics
severity critical
2026-03-16 10:47:00
Data source-managed
Import to Grafana-managed rules
New!
+ New data source-managed recording rule
No rules found.



- **[7.12] Incident Closure:** The system automatically sends a **[RESOLVED]** email once health checks pass, closing the incident loop without manual intervention.



Final Summary

This project successfully demonstrates a **Production-Grade DevOps Workflow**. By combining the power of AWS EKS with a proactive Monitoring and Alerting strategy, we have created an e-commerce platform that is not only automated but also self-aware and resilient to failures.